Thin-Pavement Thickness Estimation Using GPR With High-Resolution and Superresolution Methods

Cédric Le Bastard, Vincent Baltazart, Yide Wang, *Associate Member, IEEE*, and Joseph Saillard

***Abstract*—In the field of civil engineering, sounding the top layer of carriageways, i.e., the pavement layer, is classically performed using standard ground-penetrating radar (GPR), whose resolution is bandwidth dependent. The layer thickness is deduced from both the time delays of backscattered echoes and the known dielectric constant of the medium. This paper focuses on superresolution and high-resolution techniques, which serve to improve the time resolution of GPR signals, and presents a parametric technique and five subspace methods, namely, estimation of signal parameters via rotational invariance techniques (ESPRIT), multiple-signal classification (MUSIC) algorithm, Min-Norm, and their polynomial versions root-MUSIC and root-Min-Norm. The performance of these algorithms will be compared in terms of resolution power as well as root-mean-square error on the estimated thickness. The paper also presents the results of computer tests and radar measurements in the far field.**

***Index Terms*—Civil engineering, ground-penetrating radar (GPR), nondestructive testing (NDT), pavement survey, resolution, root mean square error (RMSE), time-delay estimation (TDE).**

I. Introduction

GROUND-PENETRATING radar (GPR) is commonly applied in the field of civil engineering as a nondestructive testing (NDT) method (e.g., see [1]–[3] and the references therein for a complete review). Within the range of centimeter wavelengths, radar techniques are more specifically dedicated to the pavement survey [1], [4]–[8]. For these applications, the roadway is assumed to be horizontally stratified. The vertical structure of roadway can then be deduced from radar profiles by means of echo detection and amplitude estimation. Echo detection provides the time-delay estimation (TDE) associated with each interface, whereas amplitude estimation is used to retrieve the wave speed within each layer. In the literature, electromagnetic inversion and layer stripping constitute two methods for estimating both of these parameters [9], [10].

When undertaking the control of small pavement thicknesses, the main difficulty with data processing lies in the echo detection of the first two backscattered echoes. This paper will focus on the TDE by assuming that the permittivity is known or has been measured and will review some of the TDE techniques that prove to be better suited for estimating thin pavement thicknesses from GPR data.

For conventional signal processing methods such as fast Fourier transform (FFT) and cross-correlation-based methods, the capability to separate two echoes is constrained by the system bandwidth [11]. For example, assuming the dielectric constant of the pavement to equal 5, according to [12] and [13], the thickness that can be measured with a conventional GPR, whose bandwidth lies between 1 and 2 GHz, must roughly be greater than 7 and 3.5 cm, respectively.

The purpose of this paper is to enhance GPR performance in measuring thin pavement layers, approximately 2 cm thick, which corresponds to the ultrathin asphalt surfacing (UTAS) within the French standard [14]. This objective necessitates either widening the operating bandwidth B, as in [4] and [5], or improving time resolution by signal processing [11], [15]–[17]. In order to maintain use of the conventional GPR system, this paper has adopted the latter solution. Processing time resolution is defined for a given frequency bandwidth B as the minimal time shift $\Delta\tau$ between two echoes that the processing is able to distinguish. The $B\Delta\tau$ product will thereafter characterize the resolution power.

It is well known that the classical FFT-based methods are able to resolve echoes for a $B\Delta\tau$ product greater than 1 [18]. With regard to UTAS, the $B\Delta\tau$ product may be less than 0.3 (0.6) for a 1-GHz (2-GHz) bandwidth radar. In comparison with the layer-stripping method [1], [9], superresolution and high-resolution signal processing techniques would thus be more appropriate for estimating the time delays of overlapping echoes. This paper proposes to assess the performance of superresolution and high-resolution algorithms for this new area of GPR applications. The main difficulty encountered in implementing these algorithms is the potentially strong correlation between echoes.

Section II provides a simplified model of backscattered radar data. Section III reviews some of the methods in the literature for TDE. We will make special emphasis on the algorithms that offer superresolution and high-resolution capabilities. In Section IV, these techniques will be applied to simulated data that represent backscattered radar data from a horizontal stratified medium in the far field. Computer tests enable determining algorithm performance and indicate the expected improvement in GPR performance. Due to multipath, we will focus, in particular, on the effect of correlation strength between echoes. Some experiments will be presented in Section V in order to demonstrate the feasibility of the UTAS thickness measurement from a conventional GPR.

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II. Backscattered Signal Model

A number of physical considerations will be discussed in this section as an introduction to the forward model used to process radar data in Section III. The roadway is assumed to be a horizontally stratified medium with K layers. A monostatic radar is positioned at nadir in the far-field condition to control the top roadway layer, i.e., the pavement, based on a sequence of vertical radar profiles (an A-scan) along the distance axis (radargram or B-scan).

Each layer of the roadway is composed of aggregates that are embedded into a homogeneous matrix [10]. The particle size is assumed to be smaller than the radar wavelength at the typical GPR frequency, e.g., a maximum particle size of 2 cm versus 9 cm for the radar wavelength within the pavement material at 1.5 GHz. Each layer can therefore be considered as a homogeneous medium, which is characterized by an effective dielectric constant and thickness [10], [19]. According to [12] and [13], pavement permittivity remains constant within the GPR bandwidth and generally ranges between 4 and 8. Conductivity is low (between 10^{-3} and 10^{-2} S/m according to the data in [20]) and renders the dispersivity of the medium negligible. Pavement thickness varies from 1 to 6 cm [14]. For UTAS pavements, the thickness decreases to 1–3 cm [14], and permittivity is equal to about 5 [12], [13].

Homogeneity of the medium is also assumed to be maintained along the scanning distance and laterally within the antenna beamwidth. Under the far-field condition, i.e., an antenna beyond the Fraunhoffer limit, these assumptions reduce the forward model to one dimension [10]. For such a simplification, a closed-form analytical expression can be derived from the solution to Maxwell's equations [21]. The received signal is composed of the K backscattered echoes, which are reflected from the medium interfaces. According to [10], [15], [16], and [22], the echoes are simply the time-shifted copies of the transmitted signal $e(t)$. Echo amplitudes s_k depend on the dielectric contrast between interfaces via the Fresnel coefficients. An additive Gaussian white noise $w(t)$ with zero mean and variance σ^2 is assumed to represent the measurement uncertainties. These assumptions on the noise are commonly used for signal processing techniques due to the simplicity of mathematical presentation. If the noise is not white, we can always use the noise whitening technique (e.g., see [34, Sec. II-C]) as far as the covariance matrix of the noise can be measured on its own. The received signal can then be written as

$$r(t) = \sum_{k=1}^{K} s_k e(t - T_k) + w(t) \tag{1}$$

where T_k represents the propagation time delay of the kth echo.

Within the frequency domain, due to the negligible dispersivity of the medium, the coefficients of reflection s_k are constant over the frequency bandwidth. The received signal is thus a linear combination of complex exponentials (cissoids) that are modulated by the radar pulse $\tilde{e}(f)$ plus an additive noise. This mathematical form is more appropriate for applying spectral analysis techniques to TDE

$$\tilde{r}(f) = \sum_{k=1}^{K} s_k \tilde{e}(f) e^{-2j\pi f T_k} + \tilde{w}(f) \tag{2}$$

where the $\tilde{\cdot}$ symbol represents the corresponding Fourier transform of the time functions in (1) and f stands for frequency. For the N discrete frequencies within bandwidth B, the received signal can be written in the following matrix form:

$$\tilde{\mathbf{r}} = \mathbf{\Lambda A s} + \tilde{\mathbf{w}} \tag{3}$$

with the following notational definitions:

- $\tilde{\mathbf{r}}$ $= [\tilde{r}(f_1)\ \ \tilde{r}(f_2) \cdots \tilde{r}(f_N)]^T$—$N \times 1$ observation vector; superscript T denotes the transpose operation;
- $\mathbf{\Lambda}$ $= \mathrm{diag}(\tilde{e}(f_1), \tilde{e}(f_2), \ \cdots \ \tilde{e}(f_N))$—$N \times N$ diagonal matrix, whose elements are the Fourier transform of the pulse;
- $\mathbf{A}$ $= [\mathbf{a}(T_1)\ \ \mathbf{a}(T_2) \cdots \mathbf{a}(T_K)]$—the mode matrix of size $N \times K$ and is composed of the vector $\mathbf{a}(T_k) = [e^{-2j\pi f_1 T_k}\ e^{-2j\pi f_2 T_k} \cdots e^{-2j\pi f_N T_k}]^T$, which is known as either the mode or steering vector of size $N \times 1$, and the vector $\mathbf{\Lambda a}(T_k)$ represents a parameterized model of the backscattered echoes;
- $\mathbf{s}$ $= [s_1\ s_2 \cdots s_K]^T$—$K \times 1$ vector that contains the amplitudes of echoes s_k;
- $\tilde{\mathbf{w}}$ $= [\tilde{w}(f_1)\ \ \tilde{w}(f_2) \cdots \tilde{w}(f_N)]^T$—$N \times 1$ vector in which each sample is a white Gaussian noise with zero mean and variance σ^2;
- f_n $= f_1 + (n-1)\Delta f$—equispaced frequency samples, where f_1 is the beginning of the bandwidth and Δf is the frequency shift.

In most instances, radar data are processed on a scan-by-scan basis without taking into account the lateral continuity of the medium [15], [16]. The distance between radar profiles is assumed to be greater than the antenna beamwidth, so that radar scans can be viewed as independent snapshots of the given data model.

This model is then applied to process either the pulse radar or the step-frequency radar. For the former, the observation vector $\tilde{\mathbf{r}}$ represents the Fourier transform of the GPR signal. The pulse shape $\tilde{e}(f)$ within matrix $\mathbf{\Lambda}$ would thus take both the transmitter (Tx) radar pulse and the frequency antenna response into account. For the step-frequency radar, the measurements from a network analyzer are derived from the observation vector, whereas $\tilde{e}(f)$ merely includes the antenna response [21]. In practice, the radar pulse is normally estimated as the backscattered signal from a metallic plane [1], [10] by assuming that the antenna response is not coupled with the dielectric property of the underlying target.

Once the TDE has been performed with the techniques presented in Section III, the data model also enables retrieving the permittivity within each layer of the stratified medium. This step would require a preliminary estimation of the source vector $\mathbf{s}$ in (3), according to the formula presented in [23]. Permittivity can then be calculated from the echo amplitude as for the layer-stripping method in [1] and [10].

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III. TDE

This section reviews three families of methods that are aimed at performing TDE based on the data model in (3). We have focused, in particular, on those families that are better suited for estimating the overlapping echoes and for accommodating the UTAS survey. Li and Wu [15] and the subsequent improvement by Wu *et al.* [16] propose an iterative algorithm, which is equivalent to the maximum-likelihood method. This algorithm offers superresolution capability and has been tested on pavement profiling up to $B\Delta\tau = 0.3$. In comparison with the latter work, we will take advantage of the data model in order to apply the TDE methods derived from the domain of spectral analysis. Some of these methods require a preliminary whitening of the data by means of the radar pulse. One possible discrepancy encountered during this operation, which is indicated in [15] and [16], is readily solved due to the special shape of the radar pulse. In addition, we will propose a more general assessment of these algorithms for pavement thickness estimation. These methods can be expressed according to the data correlation matrix (e.g., see [24, Sec. VI-E]), as defined in the following subsection. The FFT-based method will serve as a reference for assessing the enhancement of superresolution and high-resolution methods, both of which will be presented thereafter. We will briefly recall the principle behind the three main subspace algorithms that make use of the eigenstructure of the data covariance matrix to perform TDE. Some practical considerations relative to their use will be rapidly discussed as well.

A. Correlation Matrix of the Data Model

The correlation matrix is defined as an ensemble average of snapshots of the data vector $\tilde{\mathbf{r}}$ as

$$\mathbf{\Gamma} = \langle \tilde{\mathbf{r}} \tilde{\mathbf{r}}^H \rangle \tag{4}$$

where $\langle \cdot \rangle$ denotes the ensemble average, and the superscript H indicates the complex conjugate transpose. According to the data model in (3), the correlation matrix can be written in the following form:

$$\mathbf{\Gamma} = \mathbf{\Lambda} \mathbf{A} \mathbf{\Gamma}_s \mathbf{A}^H \mathbf{\Lambda}^H + \mathbf{\Gamma}_{\tilde{\mathbf{w}}}. \tag{5}$$

In (5), $\mathbf{\Gamma}_{\tilde{w}}$ and $\mathbf{\Gamma}_s$ constitute the correlation matrices of the noise $\tilde{\mathbf{w}}$ and source vector $\mathbf{s}$, respectively. Both are defined in the same manner as for matrix $\mathbf{\Gamma}$.

It is well known that the correlation coefficient between echoes ρ may exert great influence on algorithm performance (e.g., see [23]). Then, the test of algorithms in Section III will be considered with a versatile correlation magnitude in the matrix $\mathbf{\Gamma}_s$. This parameter will thus be considered in this paper through the expression of the source covariance matrix $\mathbf{\Gamma}_s$. For example, when only two echoes are included in the data model, this correlation matrix is expressed as follows:

$$\mathbf{\Gamma}_s = \begin{pmatrix} E_1 & \rho^* \sqrt{E_1 E_2} \\ \rho \sqrt{E_1 E_2} & E_2 \end{pmatrix} \tag{6}$$

where the superscript $*$ denotes complex conjugate, $E_k = \langle |s_k|^2 \rangle$, and s_k has been defined in (2).

B. Conventional Method

To estimate propagation delays, the standard processing is based on either the inverse FFT (IFFT) of the observation vector for a conventional step-frequency radar or the cross-correlation function for conventional GPR systems [10], [23]. These methods can be easily implemented and are robust with regard to signal-to-noise ratio (SNR). For both solutions, the time resolution power corresponds to the inverse of the frequency bandwidth, i.e., the product $B\Delta\tau$ is greater than 1. For the IFFT method, a weighting window improves system sensitivity and allows identifying a tradeoff between the resolution and detectability of weak echoes (e.g., see [18]). With a Hanning window for example, the resolution lies close to twice the inverse of the bandwidth, i.e., the product $B\Delta\tau$ is greater than 2, yet the variance on TDEs is reduced. This standard processing will serve as a reference in Section IV in assessing the improvements to both the superresolution and high-resolution methods, which are presented in the following subsections.

C. Superresolution Method

Superresolution methods afford an improved resolution power whose limit depends on SNR, i.e., the higher the SNR is, the better the resolution power becomes. Among these methods, the ones that are based on the autoregressive (AR) modeling of data have met a large audience in the past because they can be implemented as data series analysis technique (e.g., see [18] for a review). They have been originally proposed to estimate close frequencies of narrow-band signals [18] and, later on, to array signal processing (e.g., [26]) and to TDE (e.g., [27]).

In the case of ideal pulses, i.e., $e(t)$ being a Dirac pulse, the signal amplitude $\tilde{r}(f_n)$ in (2) can be written as a linear combination of the samples at the adjacent frequencies with additive white noise. The AR coefficients of this parametric model provide the time power density, whose peak locations yield an estimation of propagation delays [18].

Among the algorithms that are reported in the literature, the least-squares modified Yule–Walker (LSMYW) [28] method has been shown in [29] and [30] to lead to the best AR estimation. The order of the model must exceed the number of echoes K. Various criteria are proposed in the literature for determining this order, depending on both data dimension N and SNR.

D. High-Resolution Methods

In comparison with the superresolution method, multiple-signal classification (MUSIC), Min-Norm, and estimation of signal parameters via rotational invariance techniques (ESPRIT) offer three algorithms whose parameter estimation is asymptotically unbiased and resolution power is asymptotically infinite (i.e., it does not depend on both the frequency bandwidth and SNR). This family of algorithms is also known as subspace methods because they are based on the eigenstructure properties of the correlation matrix $\mathbf{\Gamma}$. These algorithms have given rise to a great number of applications in the areas of array signal processing (e.g., see [24] and [31]–[33]) and spectral

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analysis [18], [25], [34], plus a smaller number of applications for TDE (e.g., [11], [17], [23], [35], and [36]).

1) Eigenstructure of the Covariance Matrix: High-resolution algorithms are based on the eigenstructure of the data correlation matrix. Provided that the correlation matrix $\mathbf{\Gamma}$ is of full rank, and its size is greater than the number of echoes, the eigenvalue distribution enables distinguishing between the signal and the noise subspaces [24], [31]. The K eigenvectors, which are associated with the K highest eigenvalues (with the $N-K$ lowest eigenvalues), generate the signal subspace (the noise subspace). Due to the Hermitian property of the matrix, both subspaces are perpendicular to each another.

2) MUSIC Algorithm: The MUSIC algorithm consists of defining a cost function that measures the orthogonality between the noise subspace and a parameterized signal vector [24], [31], i.e., the vector $\mathbf{\Lambda a}(t)$, according to the data model in (3). For $t=T_k$, $\mathbf{\Lambda a}(T_k)$ is perfectly perpendicular to the noise subspace, which results in a null of the cost function. The TDE is then to find the maximum of the inverse of the cost function, which is also called the pseudospectrum in the literature. Due to its extensive dynamic range, a high sampling rate is required to accurately describe pseudospectrum variations and to locate the peak. This last step may induce a substantial increase in computing time.

3) Min-Norm Algorithm: For this algorithm, time delays are also determined from the magnitude of a cost function. Min-Norm requires calculating a vector that lies in the noise subspace with minimal Euclidean norm [33]. The use of this special vector to define the cost function has been shown to improve the time resolution in comparison with MUSIC [34], yet with a larger variance [37].

4) Polynomial Versions of MUSIC and Min-Norm Algorithms: Root-MUSIC and root-Min-Norm stand for the polynomial versions of MUSIC and Min-Norm algorithms, respectively [32]. They are suitable for equispaced frequency samples and therefore can be applied to the data model in (3). In comparison with the algorithms described above, root-MUSIC and root-Min-Norm require less computational resources and are faster in obtaining the TDE. Moreover, they provide the same accuracy as before while displaying an improved time resolution [37].

The search for time delays that cancel the cost function in MUSIC or Min-Norm entails searching for the complex roots $z=e^{-2j\pi\Delta ft}$ of a polynomial for root-MUSIC and root-Min-Norm. The time delays of echoes are then retrieved from the argument of the K roots: closest to the unit circle for root-Min-Norm and jointly inside it for root-MUSIC.

5) ESPRIT Algorithm: The ESPRIT algorithm was proposed in [25] for array signal processing and was then applied to TDE in [36] under the hypothesis of $e(t)$ being a Dirac pulse. In comparison with MUSIC and Min-Norm algorithms, ESPRIT provides a direct estimation of the time delays that requires less computational resources. In comparison with MUSIC, ESPRIT achieves improved variance according to [18].

The covariance matrix eigenvectors, which belong to the signal subspace, are formed into a matrix and partitioned into two overlapping subband data. Subband data usually comprise $N-1$ samples overlapped with one another by $N-2$ samples [25]. Subband data elements are related to each other by a $K\times K$ matrix. It has been shown for equispaced frequency samples that the time delays of the K echoes can be retrieved from the eigenvalues of this $K\times K$ matrix (e.g., see [36]).

E. Some Remarks on the Implementation of Algorithms

The implementation of the selected algorithms usually requires two preprocessing steps of the data. The parametric and the ESPRIT algorithms require at first to whiten the data vector $\tilde{\mathbf{r}}$ by the radar pulse shape $\tilde{e}(f)$. In addition, a special averaging technique is used to reduce the sensitivity of algorithms to the correlation magnitude between echoes.

1) Whitening the Data: According to the data model in (3), the preprocessing is formed by the data vector $\tilde{\mathbf{r}}^w=\mathbf{\Lambda}^{-1}\tilde{\mathbf{r}}$. It is pointed in [15] and [16] that the data whitening changes the behavior of noise and may degrade the SNR of the largest echoes. This discrepancy is readily circumvented for subspace algorithms by performing the general eigenvalue decomposition as originally proposed in [24]. Moreover, because the pulse of conventional radar is of Ricker type [38], the magnitude $\tilde{e}(f)$ shows smooth variations with no zero occurrence within a large frequency bandwidth.

The data whitening is specifically required before processing the data in Section V. For the computer tests in Section IV, the radar pulse is assumed to be ideal, i.e., $e(t)$ is a Dirac pulse. The performance of algorithms can then be readily compared with each other.

2) Averaging Techniques for the Estimation of the Correlation Matrix: In practice, the correlation matrix is estimated from a finite number of samples, i.e., the observation vector $\tilde{\mathbf{r}}$ over a given bandwidth [see (3)]. To ensure a reliable TDE, this number depends on SNR [11] and is found to vary between 50 and 200.

The influence of the correlation warrants a special attention within a multipath environment. The source vector $\mathbf{s}$ in model (3) represents the different paths from the same original source. Consequently, the cross correlation between echoes, i.e., the magnitude of ρ in (6), may be strong enough to upset algorithm operations. This is particularly true on simulated data and experimental data from roughless surfaces. In this situation, the correlation matrix is estimated using either the well-known spatial smoothing process (SSP)–modified SSP (MSSP) averaging technique in [23] or the newer smoothing technique (DEESE) in [39]. Both techniques show the same performances for a large amount of data, but with heavier computer loading for DEESE [39]. Therefore, only the SSP–MSSP averaging technique will be discussed in Section IV-C.

IV. Computer Tests

This section aims at illustrating using simulated data the feasibility of estimating thin pavement thickness by means of superresolution and high-resolution methods. Performance of the algorithms studied herein has been derived from both the resolution power and relative error on the thickness. Simulated data conform to the model (3), with $e(t)$ a Dirac pulse. In this section, we will consider the null and full correlation

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cases between echoes. The null case reflects the ideal situation and allows determining the best performance and limitations of the algorithms; for the full correlation case, the subband averaging technique presented in Section IV-C is required to run the algorithms. The best choice of parameters for this averaging technique will be studied next and then applied to the experimental data in the following section for which no prior knowledge can be assumed on the strength of the correlation between echoes.

A. Data Set and Testing Method

The data in model (3) consist of the backscattered signal from a two-layer medium in the far field. Algorithms are tested under the special condition when only the two first echoes are selected with a varying correlation coefficient between them. Because of the varying correlation coefficient, the algorithms are directly tested using the correlation matrix $\mathbf{\Gamma}$, which conforms to the model in (5). For the simulations, the correlation amplitude ρ in (6) is set to 0 (null correlation) and 1 (full correlation), whereas the correlation matrix of noise is estimated from independent trials of white noise.

The data set is composed of 101 equispaced frequency samples within a 2-GHz bandwidth. Over the dielectric constant range for pavement materials in [12] and [13], the dielectric constant of the medium has been set at 5 in the following simulations. SNR was defined as the ratio of the average signal power to the noise variance over the frequency bandwidth and varies between -10 dB (low SNR) and 30 dB (high SNR). To determine the limitation in resolution power, different data sets have been generated by varying the thickness from 1 mm to 20 cm such that the corresponding $B\Delta\tau$ product spans the interval [0.03, 6].

The performance of various algorithms has been established by a Monte Carlo process, which consists of 50 independent runs (U) of algorithms. For each run, the correlation matrix is estimated from 100 independent snapshots, whereby the algorithms perform an estimate of the thickness. For each snapshot, a white noise vector has been generated, with the noise being independent between snapshots. As for the subspace techniques, the signal subspace rank is equal to the expected number of echoes, i.e., 2. For the parametric technique, the order of the AR model has been optimized with regard to resolution power.

The root-mean-square error (RMSE) on the measured thickness can be used to assess the overall performance. It has been found more useful, however, to calculate the relative error (RRMSE), as defined in the following formula:

$$\text{RRMSE}\ (\%) = 100 \times \frac{\sqrt{\frac{1}{U}\sum_{i=1}^{U}(\hat{e}_i - e)^2}}{e} \qquad (7)$$

where $\hat{e}$ denotes the estimated thickness and e is the true value. A performance is considered to be reasonably good whenever the RRMSE lies below 5%. Simulations include MUSIC, Min-Norm, ESPRIT, and LSMYW. To lighten the computer load, only the polynomial versions of MUSIC and Min-Norm have been tested. We will now provide the results from the computer tests for the two extreme correlation coefficients.

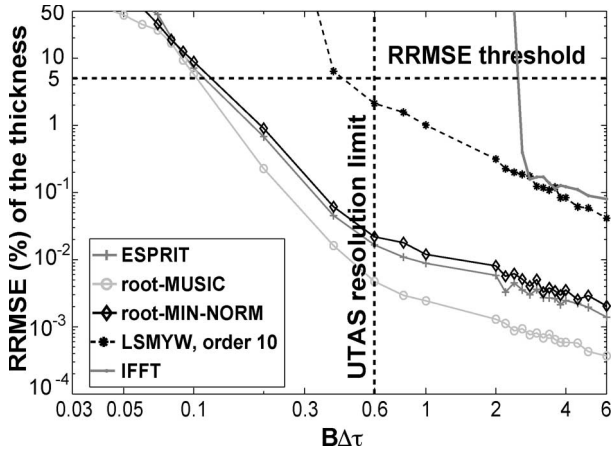


Fig. 1. RRMSE variations versus $B\Delta\tau$ product for the four algorithms in the situation of ideally uncorrelated echoes; size of the signal subspace, $K = 2$ (root-MUSIC, root-Min-Norm, ESPRIT); AR order $= 10$ (LSMYW); SNR $=$ 20 dB.

B. Null Correlation Case

This section examines the ideal situation when the first two echoes are totally uncorrelated, which allows determining the best performance and limitations of the algorithms. Fig. 1 shows relative RMSE variations with respect to $B\Delta\tau$ at high SNR levels (20 dB). The resolution power is said to be improved when the $B\Delta\tau$ product decreases for a given RRMSE threshold. The vertical and horizontal dashed lines show the two thresholds for fitting with the UTAS survey application: $B\Delta\tau < 0.6$ (for a 2-GHz GPR bandwidth) and RRMSE $< 5\%$.

We are able to observe that RRMSE continuously decreases with the increase of $B\Delta\tau$ product. At RRMSE $= 5\%$, the three subspace algorithms roughly lead to the same ultimate resolution, i.e., the corresponding $B\Delta\tau$ product equals 1/10. The ultimate resolution of the parametric algorithm is about four times greater, whereas that of the IFFT is beyond 2 as expected. Resolution performances of the different algorithms are rated as root-MUSIC, root-Min-Norm and ESPRIT (which both produce very similar performances), followed by LSMYW, and IFFT at last.

For the algorithms that can fulfill the UTAS criterion, Fig. 2 shows the ultimate resolution variations (i.e., $B\Delta\tau$ product that corresponds to 5% in RRMSE) with regard to SNR. The resolution power decreases (i.e., $B\Delta\tau$ gets larger) with decreasing SNR for the three subspace algorithms. Root-MUSIC still offers the best performance versus the root-Min-Norm and ESPRIT algorithms. The subspace algorithms, however, tend toward similar performance levels at higher SNR values. Two thresholds have been drawn as horizontal dashed lines for the $B\Delta\tau$ product in order to take account of the versatile GPR bandwidth, i.e., $B = 1$ or 2 GHz. The minimum SNR to be satisfied for the UTAS survey application thus depends on both the bandwidth and the algorithms. For a 2-GHz bandwidth, the minimum SNR equals -7 and -3 dB for root-MUSIC and

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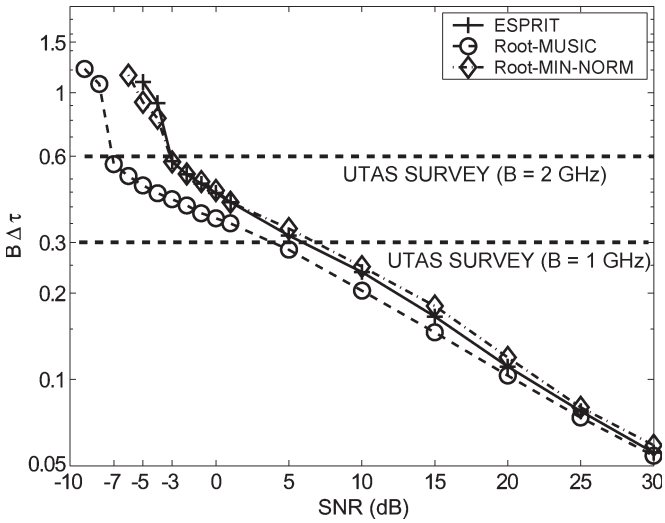


Fig. 2. Ultimate resolution variations versus SNR for the subspace algorithms in the situation of ideally uncorrelated echoes; size of the signal subspace, $K = 2$.

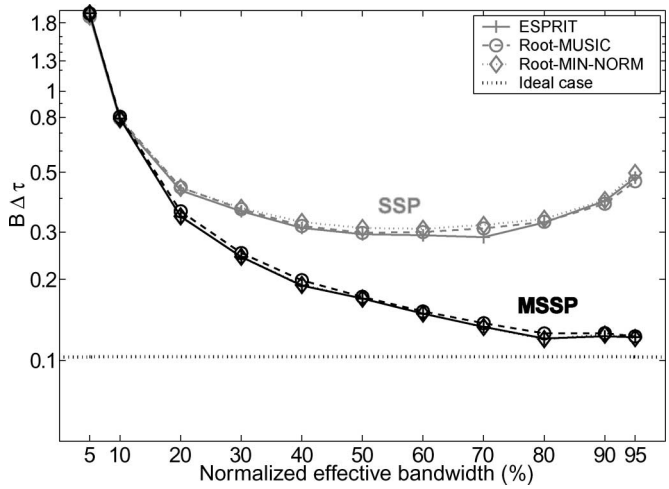


Fig. 3. Ultimate resolution variations versus the effective bandwidth with the forward (SSP) and forward–backward (MSSP) frequency smoothing techniques; SNR $= 20$ dB.

both root-Min-Norm and ESPRIT, respectively. A higher SNR is required to satisfy the UTAS survey when $B\Delta\tau$ decreases.

C. Full Correlation Case

In this case, the rank of the source correlation matrix is less than the number of echoes, which results in failure of the subspace algorithms. To alleviate this problem, [23] adapted the spatial smoothing technique proposed in [40] to the TDE problem. The frequency bandwidth (N data) has been organized into M overlapping subbands of L data each. L will be referred to below as the effective bandwidth. For a maximum overlapping ratio between subbands, L, N, and M are related to each other as follows:

$$N = L + M - 1. \tag{8}$$

The frequency smoothing technique consists of averaging the M correlation matrices calculated from each subband vector (L data); its application is also called the forward technique (SSP) and the forward–backward technique (MSSP) if the data are averaged in the direct order (as SSP) and the reverse order simultaneously. Consequently, the correlation matrix to be processed by the subspace algorithms is reduced to the size $L \times L$.

The relations in [23] show that the reduction in correlation magnitude primarily depends on the number M of subbands. As M increases, the size of subbands L decreases, as does the resolution power of the algorithm. A tradeoff between correlation magnitude and resolution power must therefore be made.

In comparison with the literature, this subsection is aimed at precisely identifying the best parameters for the smoothing technique. Fig. 3 presents the ultimate resolution variations (i.e., the $B\Delta\tau$ product at RRMSE $= 5\%$) versus the normalized effective frequency bandwidth, i.e., L/N ratio, at SNR $=$ 20 dB. The lowest $B\Delta\tau$ product yields the optimum achievable performance, from which both the optimum number M and the size of the subbands L can be derived. For comparative purposes, the horizontal dashed line shows the optimum performance obtained in Fig. 2 for root-MUSIC at SNR $= 20$ dB. We first observe that the results from each averaging technique do not depend on the algorithm employed. Second, the forward averaging SSP technique leads to a minimal $B\Delta\tau$ product when the effective frequency bandwidth lies between 50% and 70%. We obtain enhanced performance with the forward–backward MSSP technique and an effective frequency bandwidth between 80% and 90%, which implies that the effect of reversing data order in the MSSP averaging technique has the greatest influence in comparison with the number of overlapping subbands. Moreover, the MSSP technique allows approximating the optimum performance shown in Section IV-B, where echoes are completely uncorrelated.

V. Experimental Results

A. Experimental Device

The performances of the studied algorithms have been tested in the laboratory on roughless materials. A GPR (GSSI SIR-3000) has been used for the experiments and was combined with a bistatic device whose Tx and receiver (Rx) are very close to each other (see Fig. 4). The antennas for both Tx and Rx devices are of the bow-tie type, and the back lobes of the antennas have been reduced by a top shielding in each device. The central frequency is 1.7 GHz in air, and the bandwidth is equal to about 2 GHz at -6 dB. The GPR was raised above various slabs of materials to insure both vertical incidence and far-field conditions. We are well aware that the bow-tie antenna is not appropriate for far-field measurements because of its wide radiating pattern.

These tests have been performed on a polished granite slab of 10.4 cm thick and on a 2-cm-thick polyvinyl chloride (PVC) slab set on a metallic plane. The dielectric constant of both materials has been measured at 2 GHz with the microwave probe presented in [12] and [13] and is equal to $5.07 + 0.15j$ and $2.9 + 0.015j$, respectively. Both materials therefore

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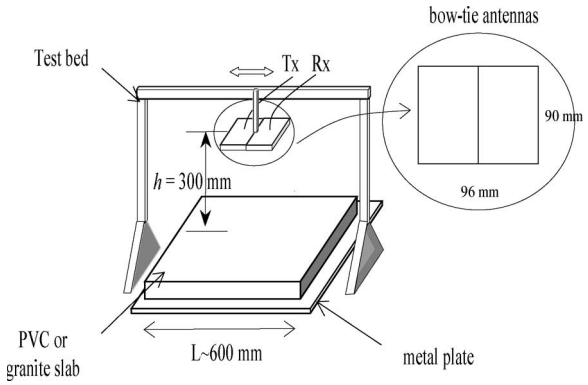


Fig. 4. Scheme of the experimental device with GPR.

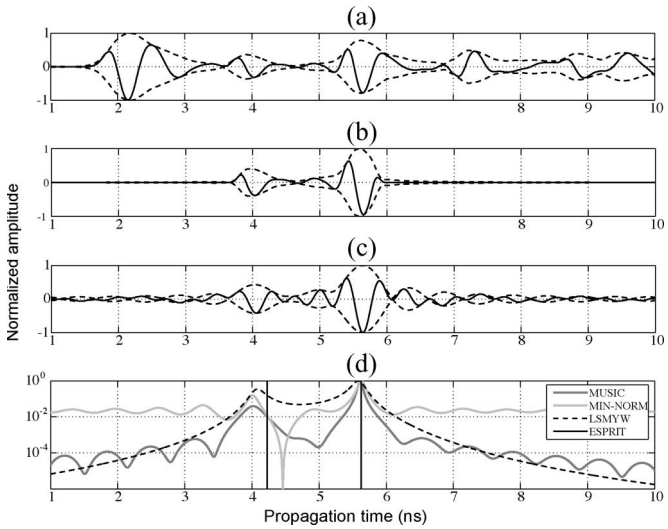


Fig. 5. Processing of backscattered echoes from the granite slab (resolved echoes, $B\Delta\tau = 3$). (a) Raw GPR data. (b) Time-filtered data. (c) Whitened data. (d) Pseudospectrum from the MUSIC, Min-Norm, LSMYW, and ESPRIT algorithms. GPR data on panels (a)–(c) have been plotted in both (solids lines) amplitude and (dashed lines) magnitude.

correspond to $B\Delta\tau \approx 3$ and $B\Delta\tau \approx 0.45$, respectively, and both experimental settings enable testing the algorithms in situations with resolved and overlapping echoes, respectively.

PVC has been used instead of a pavement material since polishing of the pavement surface was not affordable at the time of these experiments. A PVC slab may be equivalent to a 1.5-cm-thick pavement slab, in assuming that the corresponding dielectric constant is roughly equal to 5 [12], [13]. Using a PVC material instead of pavement is expected to reduce multipath occurrence and to raise the correlation magnitude between echoes. Due to a smaller dielectric contrast, this modification is also expected to lower the SNR of echoes by a few decibels (approximately -3 dB). Consequently, the results on a pavement material are expected to be slightly better than those on PVC.

B. Data Set

For each snapshot, we set a 10-ns range time and chose 1024 as the number of samples for a sampling frequency of 102 GHz. The antenna was moved slightly between various snapshots to generate independent spatial measurements. Figs. 5(a) and

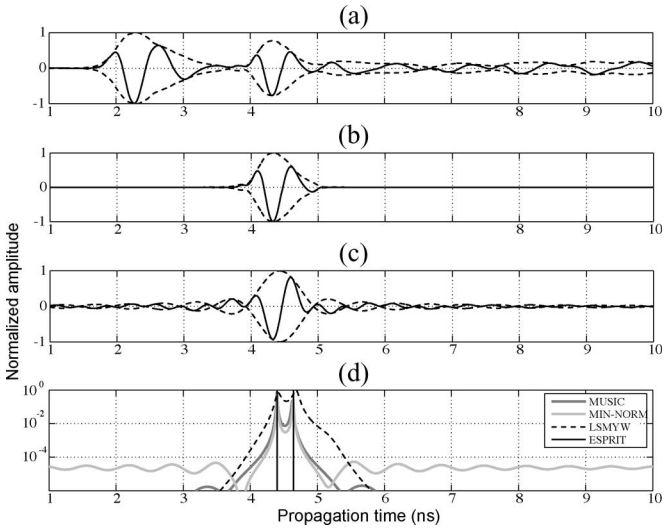


Fig. 6. Processing of backscattered echoes from the PVC slab (overlapping echoes, $B\Delta\tau = 0.45$). See Fig. 5 for details.

6(a) display the raw experimental data; the SNR level is about 20 dB. The dashed lines represent the envelope of received signals after application of the Hilbert transform. The first echo corresponds to the airwave between the Tx and Rx devices. The second and third echoes in Fig. 5(a) correspond to the two backscattered echoes from the two first interfaces. In Fig. 6(a), these same echoes are overlapping; in the end, multiple echoes appear.

C. Preprocessings of the Data

Before employing the high-resolution and superresolution methods, two processes are performed on the data in order to bring the experimental conditions closer to the simulated situations in Section II.

1) Time Filtering: The first processing operation consists of a time filtering to retain just the two backscattered pulses due to the two interfaces. The multiple echoes and direct wave in air are then filtered, as shown in Figs. 5(b) and 6(b).

2) Data Whitening: The second operation consists of whitening the signal, for which the radar pulse is measured as the backscattered signal from a metallic plane, which has been positioned over the material under test. Finally, a chirp-Z transform is performed to obtain 101 frequency data elements within the 2-GHz bandwidth. Figs. 5(c) and 6(c) show the processed time data series to be used by the parametric and subspace methods.

3) Subband Smoothing: The correlation matrix has been estimated from a set of 20 snapshots. In both cases, we have used the forward–backward (MSSP) technique to reduce the correlation effect. The effective frequency bandwidth is chosen to be 80% of the total bandwidth, i.e., $L = 81$, according to the results shown in Fig. 3.

D. Results for Echo Detection

For subspace algorithms, the signal subspace dimension has been set as the true number of expected echoes, i.e., $K = 2$. For the parametric method, the order of the AR model has been

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TABLE I
RELATIVE BIAS IN PERCENT ON THE ESTIMATED THICKNESS OF BOTH THE GRANITE AND THE PVC SLABS

	Granite slab (ϵ = 5.07+0.15j, e = 10.4 cm, $B\Delta\tau$ = 3)	PVC slab (ϵ = 2.9+0.015j, e = 1.52 cm, $B\Delta\tau$ = 0.45)
MUSIC	2.7	0.4
root-MUSIC	2.8	0.6
Min-Norm	4.4	4.4
root-Min-Norm	4.4	4.2
ESPRIT	9.6	5.7
LSMYW	3.1	15.6

chosen to ensure obtaining the best results: this order would be equal to 4 for the resolved echoes and 10 for the overlapping echoes.

Figs. 5(d) and 6(d) illustrate the results obtained by the algorithms on the granite and PVC slabs. For visualization purposes, only the results of four algorithms are plotted. Table I lists the entire set of results, which include the polynomial versions of the MUSIC and Min-Norm algorithms. For the resolved echoes (i.e., granite slab), all methods accurately locate the echo time delay. ESPRIT reveals a larger relative error, however, which may be caused by the imperfect whitening technique used. The second case on the PVC slab illustrates the feasibility of measuring a 2-cm-thick pavement material with a 2-GHz GPR bandwidth. As opposed to conventional methods such as IFFT, the parametric and subspace methods are able to separate overlapping echoes. LSMYW, however, displays an excessive error to fit the UTAS criteria. For subspace algorithms, the relative bias is slightly higher in the case of the granite slab. The higher conductivity of granite may result in greater pulse dispersion, which has not yet been taken into account in signal model (3). ESPRIT seems to be more sensitive to dispersion phenomena.

VI. CONCLUSION

In this paper, we have performed measurements of thin pavement thicknesses using a standard GPR with a bandwidth of between 1 and 2 GHz. We have also shown that the selected signal processing techniques are able to improve the time resolution of conventional GPR. The resolution power of the high-resolution signal processing techniques varies with respect to SNR. For a high SNR value, these algorithms are able to separate overlapping echoes whose $B\Delta\tau$ product is about 1/10 at the least. For the estimation of thin pavement thicknesses, the subspace and parametric methods are able to achieve these performance levels at an SNR of greater than -7 dB for a 2-GHz GPR bandwidth. In any event, subspace methods perform better than the parametric method. For completely correlated echoes, subspace algorithms require the use of frequency averaging techniques for reducing the correlation between echoes. Two averaging techniques have been presented herein; moreover, the size and the number of frequency subbands have been studied through simulations to identify the most attractive tradeoff. The forward–backward averaging technique offers the best performance, which may approximate the perfectly uncorrelated situation. Experiments have demonstrated the importance of taking dispersion into account in the signal model and of assessing during subsequent works the influence of pavement roughness (within the Rayleigh criterion) on the correlation magnitude between echoes. The test of subspace algorithms will be extended to the amplitude estimation of echoes in order to compare overall performances with some typical pavement profiling methods such as layer stripping [10].

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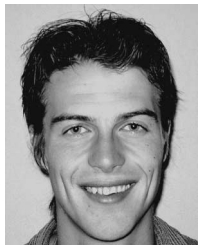
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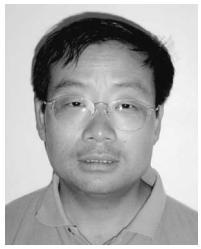
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Dr. Saillard is the organizer of the Proceedings of International Workshop on Radar Polarimetry (JIPR'90, JIPR'92, JIPR'95, and JIPR'98) and is a member of the Electromagnetic Academy.

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